

CLOUD COMPUTING COURSE

Chapter4-Cloud Storage

ABSTRACT

This chapter Introduces cloud storage systems and storage virtualization concepts.

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Storage Virtualization Definition

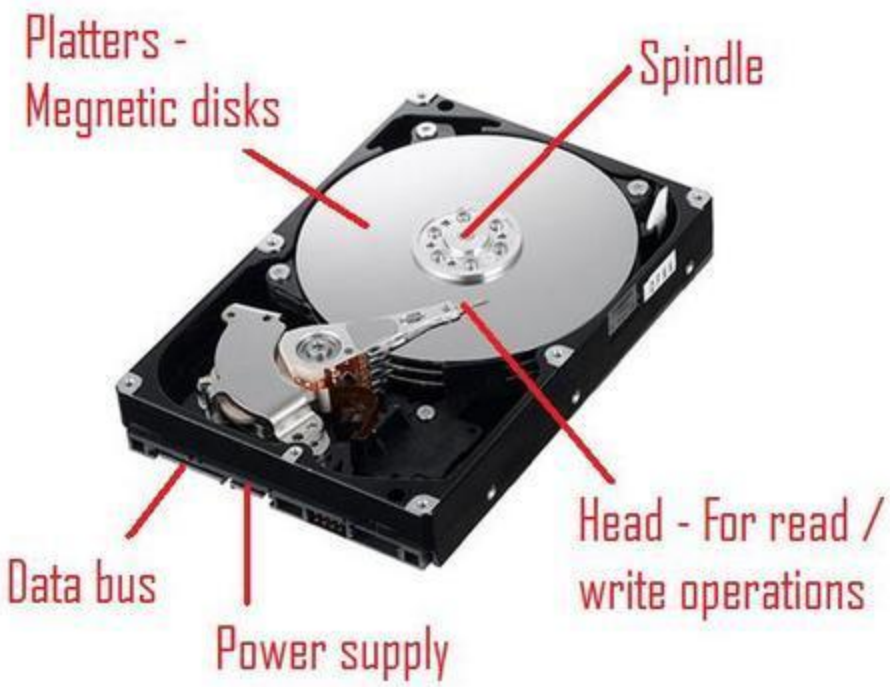
Storage virtualization is the process of grouping and sharing the physical storage from multiple network storage devices so that it looks like a single storage device, which simply managed from a central console.

When you buy a server or storage array these days, you often have the choice between different kinds of hard drives:

Mainstream Physical Disk Types

Type1: HDD

HDDs (Hard Disk Drive): are considered as an irreplaceable and important storage device that uses a **mechanical arm** with a **read/write head** to move around and read information from the right location on a **storage platter**.



HDD Types:



Currently, SATA and SAS hard drives are commonly used and **distinguished by interface** type.

1. **PATA (Parallel advanced technology attachment):** this transfers data in parallel mode and it is not used any more.

2. **SATA hard drive (Serial advanced technology attachment):** it replaced all PATA interfaces, SATA transfers data in serial mode. One of its distinctive features is that SATA is faster than PATA and have a very high capacity.



SATA HDD

3. **SAS**

   Serial attached SCSI (SAS), that is, small computer system interface. Similar to SATA, SAS uses serial transfer to achieve higher performance. Compared with a SATA hard drive, a SAS hard drive has a higher read/write speed, higher price, and **smaller c**apacity.



SAS HDD

In reliability, SAS disks are an order of magnitude safer than either NL-SAS or SATA disks. This metric is measured in *bit error rate* (BER), or how often bit errors may occur on the media.

4. NL-SAS

Near Line SAS (NL-SAS) is a type of hard drive between SATA and SAS. Therefore, an NL-SAS hard drive is developed, which consists of SAS interfaces and SATA platters (head, media, and rotational speed).



NL_SAS HDD

NL-SAS is slower than SAS.

NL-SAS is faster than SATA.

NL-SAS has the capacity and price of the SATA hard drive and the reliability of the SAS hard drive.

Type2: SSD (Solid-State Drives)

- SSDs offer up to 100 time's greater throughput than traditional hard drives, which means **faster boot ups and better overall system performance**.
- Unlike SATA and SAS hard drives, SSDs **have no moving parts** making them the most reliable and efficient option.
- The lack of moving parts also means a **reduced risk of failure** and an **increase** in **power efficiency** compared to traditional hard drives.
- SSDs, as storage media with **higher read/write speeds** than traditional HDDs, have received widespread attention.
- Traditional HDDs are disk drives and data is stored in disk sectors. The common storage medium of an **SSD is flash memory**.
- Because of the performance, speed, and reliability benefits, solid-state drives are generally **more expensive** than SAS and SATA hard drives.
- SSDs are ideal **for high frequency immediate transactional data like database, CRM or bank transactions.**
- SSDs are one of the most essential storage devices in the **ultra-thin laptop** and tablet fields.
- In addition, SSDs have distinctive features **such as shock resistance, small size, no noise, and low cooling requirements.**
- An SSD consists of **a flash controller and memory chips.** The flash controller controls the coordination of the data read/write process and the memory chips are responsible for storing data.A most common type is to use **a flash memory chip as the storage medium**, and the other type is to use **a dynamic random access memory** (DRAM) chip as the storage medium.

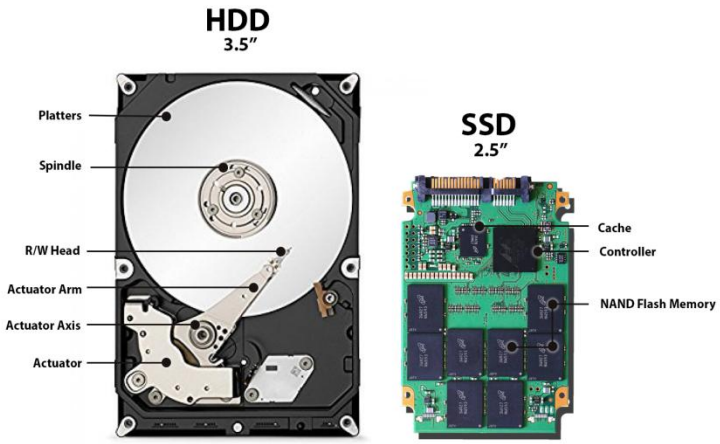
Flash-based SSDs



The most common SSDs use a flash memory chip as the storage medium. Flash memory chips can be manufactured into various electronic products, such as SSDs, memory cards, and USB drives. These devices have **small size and easy to use.**

Disadvantage of SSD

- The cost of SSDs is far higher than that of traditional HDDs.
- Less capacity than HDD
- Please watch the video in the URL: https://youtu.be/GN4YJaRA9Ss



Comparison between different physical disk types

Item	SSD	SAS	NL-SAS	SATA
Performance	Very high	High	Relatively high	Relatively high
Reliability	Minor	High	Relatively high	Minor
Price	Very high	High	Relatively low	Low
Power consumption	Minor	High	Relatively low	Relatively low
Recommended scenario	High frequency immediate transactional data like database or bank transactions.	Suitable for users with discrete data access	Suitable for backup and data archiving (low data access)	Because of the high storage capacity they offer, SATA drives are commonly used for file sharing, email, web, backup, and archival data.

Centralized Storage

Centralized storage means that all storage resources are centrally deployed and are provisioned over a unified interface.

With centralized storage, all physical disks are centrally deployed in disk enclosures and are used to provide storage services externally through the controller. Centralized storage typically refers to disk arrays.

A disk array combines multiple physical disks into a single logical unit. Each disk array consists of one controller enclosure and multiple disk enclosures. This architecture delivers an intelligent storage space featuring **high availability**, **high performance**, and **large capacity.**

Disk Array→



Based on the technical architecture, centralized storage can be categorized as:

DAS, NAS and SAN.

1. ### DAS (Direct attached storage):

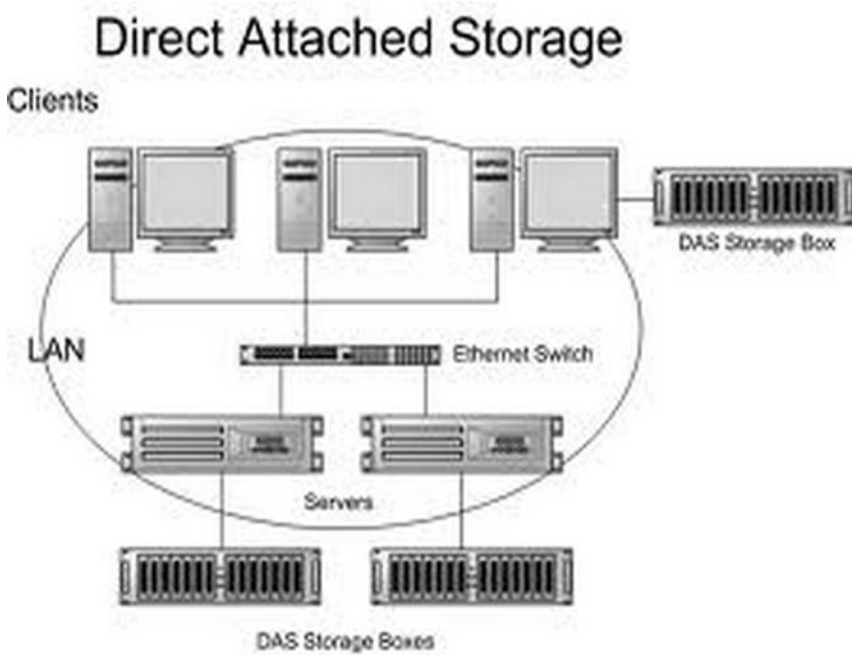
   It is digital storage directly attached to the computer accessing it.

   Examples of DAS include hard drives, solid-state drives, optical disc drives, and storage on external drives. DAS is the most widely used storage system before SAN is introduced.



DAS Types:

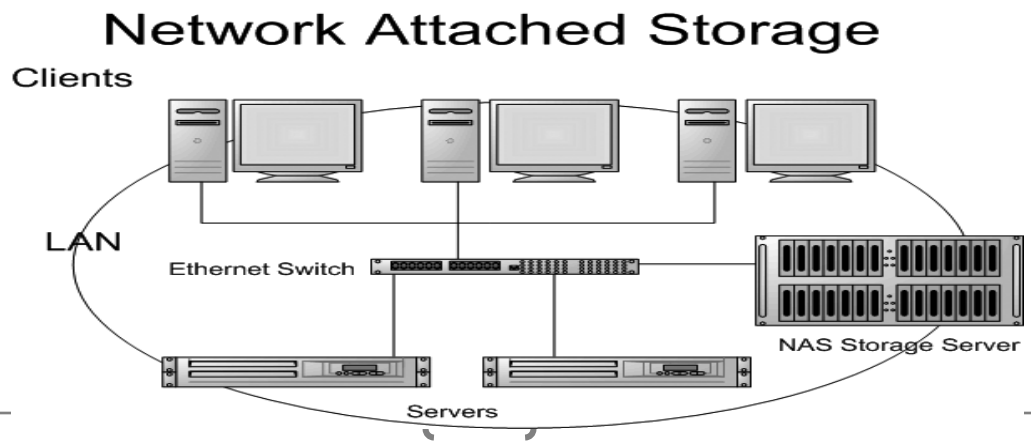
- Standalone
- JBOD (Spanned)
- RAID



2. NAS (Network attached storage)

A technology integrates distributed and independent data into a large-scale and centralized data center for access by different hosts and application servers.

- NAS is **a file-level computer** data storage connected to a computer network providing data access to heterogeneous group of clients.
- An NAS server **contains storage devices**, such as disk arrays, CD/DVD drives, tape drives, or portable storage media.
- An NAS server delivers an embedded **OS to share files** across platforms.
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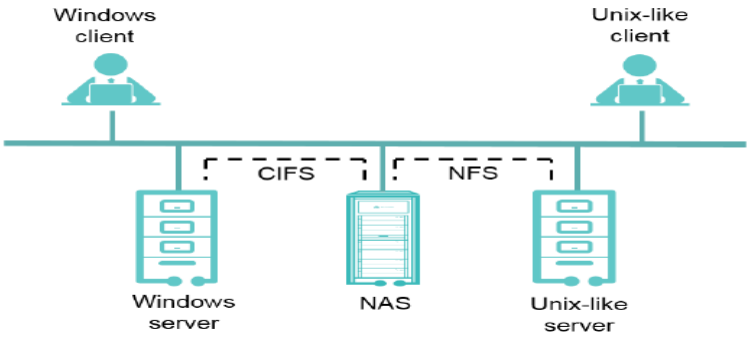


Simply speaking, a NAS is a device that is connected to the network with file storage function. Therefore, NAS is also called a **network file storage device.**

- It is often manufactured as a computer appliance – a purpose-built specialized computer, It delivers centralized file storage and management and separates storage devices from servers, **releasing bandwidth and improving performance.**

Essentially, NAS is a storage device rather than a server.

- The advantage of NAS is that it can deliver file storage services in a **fast and cost-effective** manner using existing resources in the data center. The current solution is compatible between UNIX, Linux, and Windows OSs, and can be easily connected to users' TCP/IP networks. The following shows the NAS system.



CIFS vs NFS are protocols developed to permit a client system to access the file system on remote computing devices like server or personal computers **(File Sharing Protocols).**

1. **NFS** is the "**Network File System**" specifically used for **Unix and Linux operating systems** systems to share files and folders.. It allows files communication transparently between servers and end users machines like desktops & laptops. NFS uses client- server methodology to allow user to view read and write files on a computer system.

2. **CIFS** is abbreviation for "**Common Internet File System**" used by Windows operating systems for file sharing. CIFS also uses the client-server methodology where a client makes a request of a server program for accessing a file .The server takes the requested action and returns a response. CIFS is an open standard version of the Server Message Block Protocol (SMB) developed and used by Microsoft and it uses the **TCP/IP protocol.**

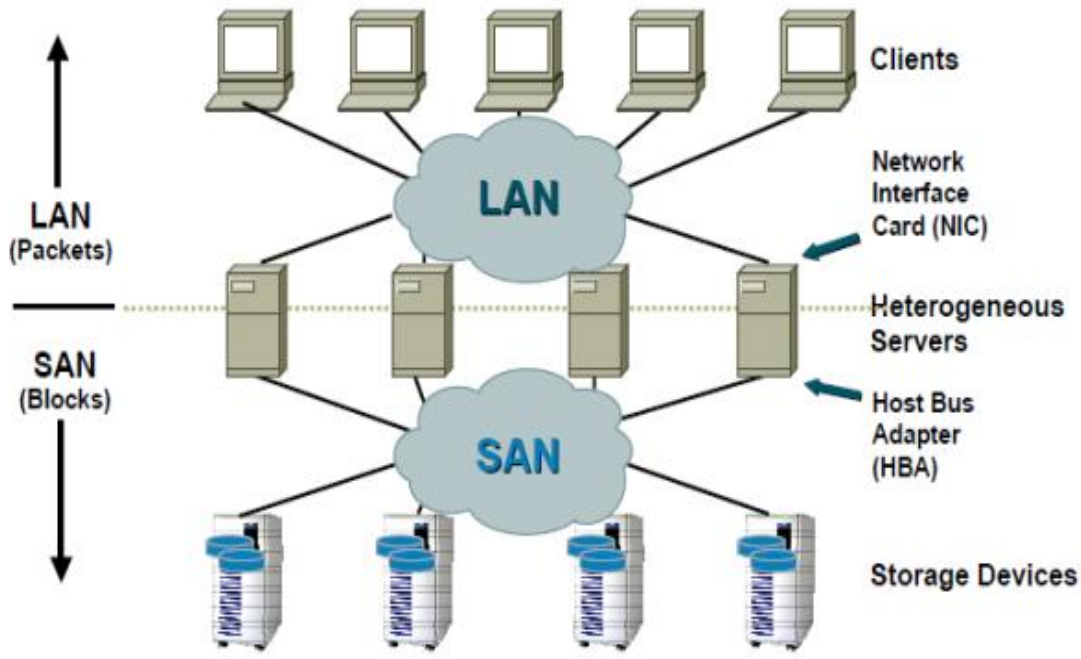
3. SAN (Storage Area Network):

Now, data has become the most valuable asset since the Internet and e-commerce are growing at an explosive rate. How to store, protect and manage data in an effective way is a big challenge for IT technicians. Here, SAN, storage area network, has been developed to provide a creative model for data storing in the data center.

***SAN Definition*:** is a specialized, high-speed network that provides block-level network access to storage.

***SAN Component*:** SANs are typically composed of **hosts, switches** and **storage devices** that are interconnected using a variety of technologies, topologies, and protocols.

SAN Access Mechanism



SAN Is a Dedicated Network for Attaching Servers to Storage Devices

SAN makes a network of storage **devices accessible to multiple servers**.

SAN devices **appear to servers as locally attached drives**, eliminating traditional network bottlenecks.

A SAN is block-based storage, leveraging a high-speed architecture that connects servers to their **logical disk units (LUNs)**.

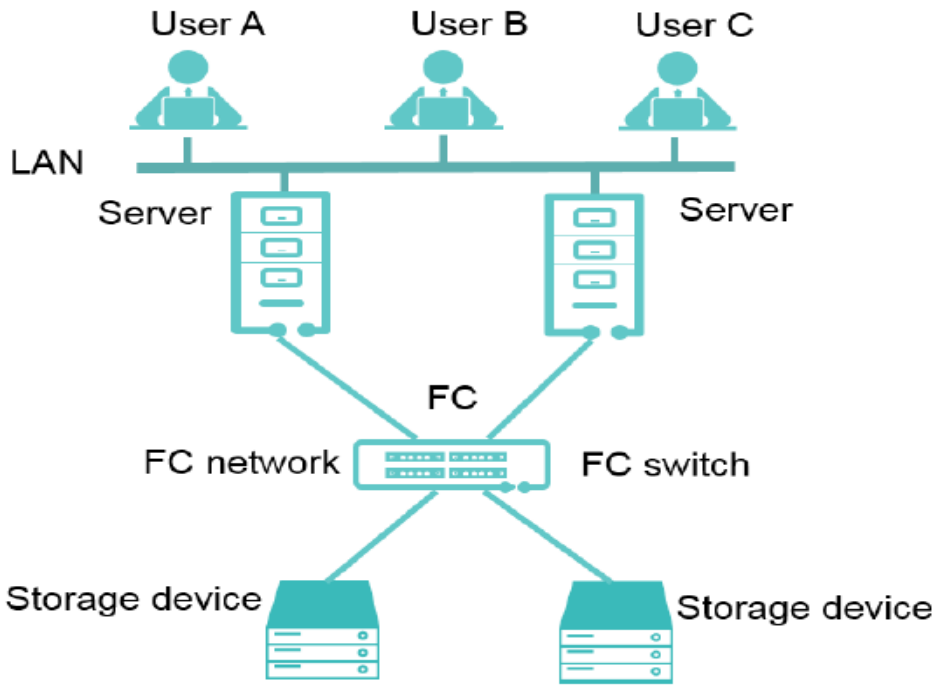
A LUN is a range of blocks provisioned from a pool of shared storage and presented to the server **as a logical disk**.

The **server partitions and formats those blocks**, typically with a file system, so that it can store data on the LUN just as it would on local disk storage.

SAN Types

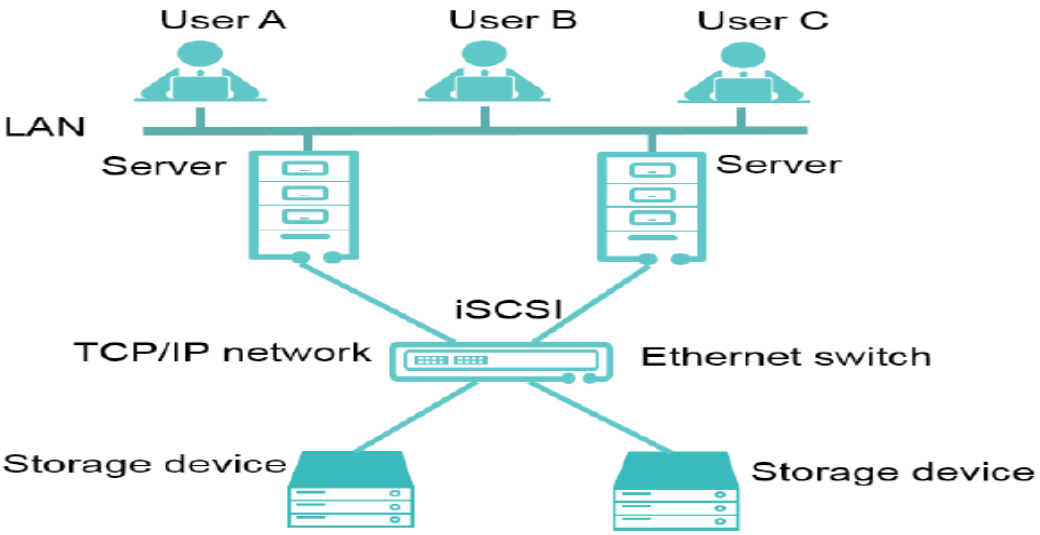
SAN solutions are available as many types, the most common two types are:

- **FC-SAN (Fiber Channel SAN):** Storage and servers are connected via a high-speed network of interconnected fiber channel switches. This is used for mission-critical applications where uninterrupted data access is required.



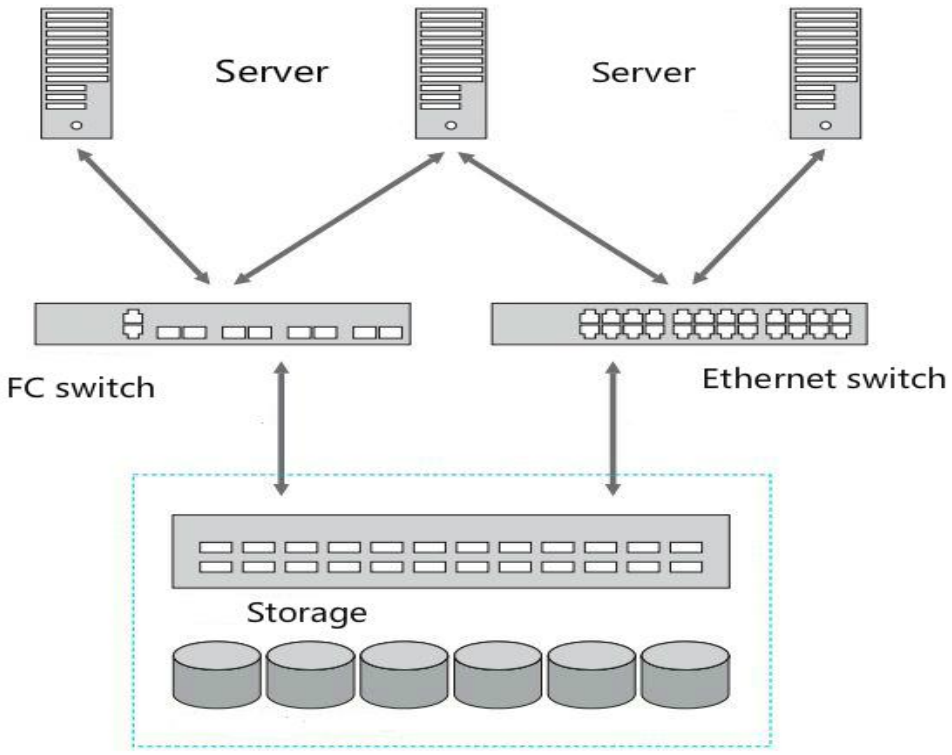
FC SAN Network

- **Internet Small Computer System Interface (iSCSI) Protocol/IP SAN**: This infrastructure gives the flexibility of a low-cost IP network. It uses iSCSI protocol, a transmission standard over TCP/IP, to transfer block data over an Ethernet network. IP SAN allows different servers to access pools of the shared block storage devices by storage protocols.



IP SAN network

Composite Solution (both FC SAN and IP SAN)



IP SAN vs FC SAN: What Is the Difference?

Here is a table shows the differences of IP SAN vs FC SAN.

	IP SAN	FC SAN
Network	Gigabit Ethernet network	fiber channel network
Cost	Low	High
Transmission speed	Fast	Extremely fast
Transmission protocol	TCP/IP	fiber channel
Extensibility	More easier to extend	Easy to extend

SAN vs. NAS

	NAS	SAN
Main Target	Are methods of managing storage centrally and sharing that storage with multiple hosts (servers) .	
Network	Ethernet-based	Ethernet and Fibre Channel
Focus on	Ease of use, manageability, scalability, and lower total cost of ownership (TCO).	high performance and low latency
Partitioning	NAS **storage controllers** partition the storage and then own the file system	The **server** partitions and formats SAN storage blocks with its file system
Protocols	CIFS, NFS	FCP, iSCSI Protocol

RAID

RAID (Redundant Array of Inexpensive Disks or Drives, or Redundant Array of Independent Disks): A data storage virtualization technology that combines multiple physical disk drive components into one or more logical units for the purposes of data reliability, performance improvement, or both.

Benefits of RAID

RAID allows us to combine drives, increase speed and improve reliability so we can store more data while our data is stored safer.

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RAID allows us to bundle multiple hard-drives that your Operating System (Windows, MacOS X, Linux, etc) will see as one single big drive.

Example

As an example, based on Windows and 3 hard-drives of say 1 Tb:
Without RAID, taking Windows as an example, 3 drives could be seen as the drives C:, D:, and E: of each 1Tb (if each drive would have one partition).
With RAID, these 3 drives can be combined to a single hard-drive, C:, with a capacity of 3Tb (depending on the RAID configuration you choose).

Where can you find RAID?

RAID can be found in any computer, server, external disk array or NAS (Network Attached Storage), where the "controller" can actually be hardware or software.

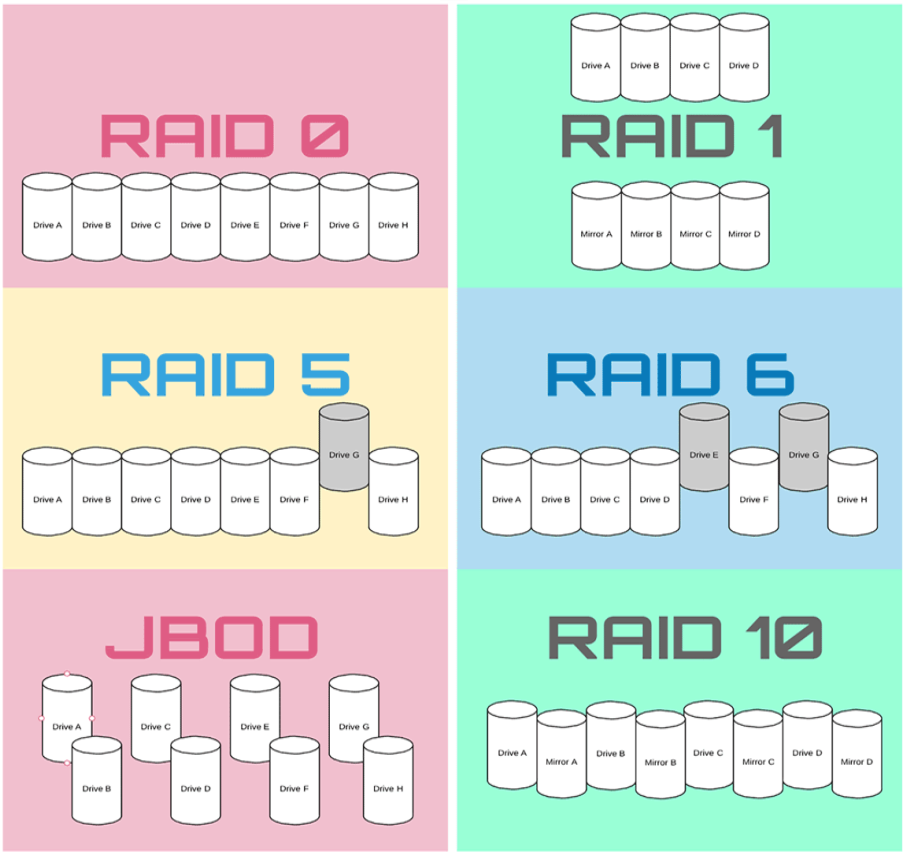
RAID Types (Variations)

RAID is an umbrella name for different variations of combining hard-disks that would then appear to your computer as one single hard-disk.
Multiple disks are combined with different goals in mind, hence the different variations. Goals can be **Faster access to data**, more **storage space** and/or **higher reliability.**
These variations are named "**RAID**" followed by a **number**. For example **RAID 0**, **RAID 1**, etc. Each fitting for a particular purpose.

The different variations come with different levels of:
Speed, **reliability**, **availability** and the **minimum number of hard-drives** needed.

RAID Variations (Types)

- RAID 0 – Striping
- RAID 1 – Mirroring
- RAID 2 – Bit level Striping
- RAID 3 / RAID 4 – Striping with Parity
- RAID 5 – Striping with rotating parity
- RAID 6 – Striping with double rotating parity
- Nested RAID (RAID 10)



RAID 0 – Striping

In RAID 0 data is written in **parallel** to 2 or more disks. Think of it as writing block 1 to disk 1, block 2 to disk 2, block 3 to disk 3 etc.

Minimum number of required drives: 2 drives (even number).

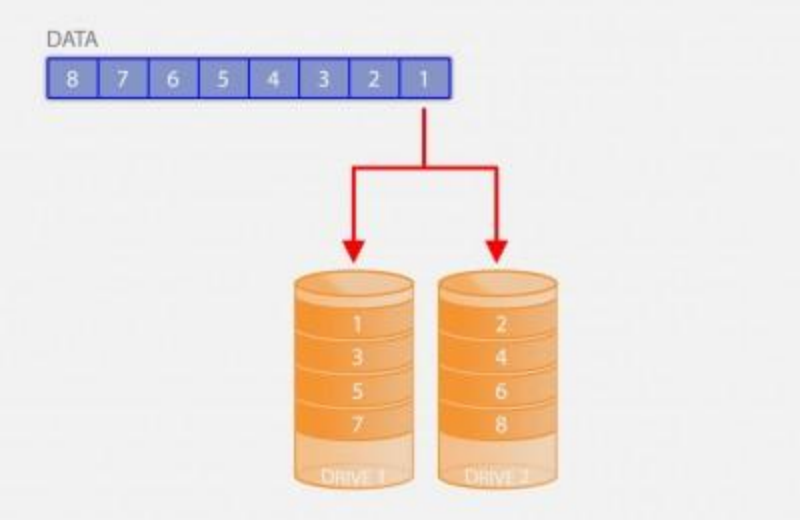
Space efficiency: 100% of the sum of the capacity of the drives.

Positives:

- The great advantage is **SPEED** as data is written or read in parallel.
- Another advantage is that all of the **storage capacity** will be available for storage.

Negatives:

The big downside is **lack of reliability**: If one disk crashes all data is gone, as half of the information will be lost. RAID 0 is not fault-tolerant.



RAID 0 – Striping

Ideal use

RAID 0 is ideal for non-critical storage of data that have to be read/written at a high speed, such as on an image retouching or video editing station.

RAID 1 – Mirroring

RAID 1 arrays mirrored data of one drive to another drive. Therefore, when your computer writes data to the array, the controller will actually write the data twice: to disk 1 and to disk 2.

If a drive fails, the controller uses either the data drive or the mirror drive for data recovery and continues operation.

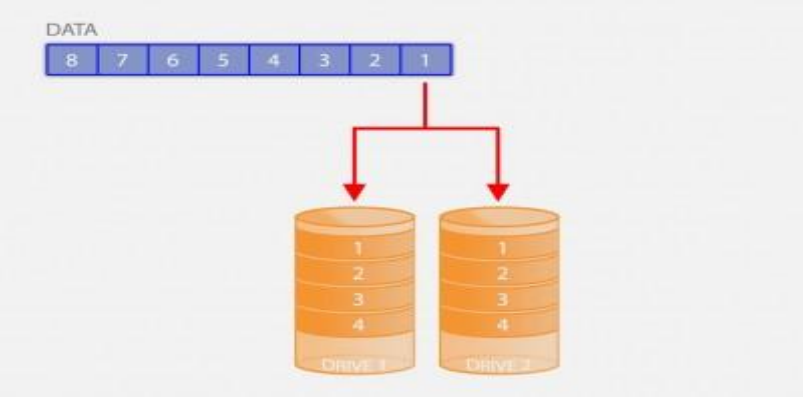
Minimum number of required drives: 2 drives (even number).

Space efficiency: 50% of the sum of the capacity of the drives.

Positives:

- Increased **reliability** since data is always saved in duplicate on two different drives. If one drive dies, the other will still be able to provide you the data. In case a drive fails, data do not have to be rebuild, they just have to be copied to the replacement drive.
- RAID 1 is a very simple technology.

Negatives:
The main disadvantage is that the effective storage capacity is only half of the total drive capacity because all data get written twice.



RAID 1 – Mirroring

Ideal use

RAID-1 is ideal for mission critical storage, for instance for **accounting systems**.

RAID 5 – Striping with rotating Parity

- RAID 5 is the most common secure RAID level.
- It requires at least 3 drives but can work with up to 16.
- Data blocks are striped across the drives and on one drive a parity checksum of all the block data is written.
- The parity data are not written to a fixed drive, they are spread across all drives, as the drawing below shows.
- Using the parity data, the computer can recalculate the data of one of the other data blocks, should those data no longer be available.

- That means a RAID 5 array can withstand a single drive failure without losing data.
- Although RAID 5 can be achieved in software, a hardware controller is recommended. Often extra cache memory is used on these controllers to improve the write performance.

Minimum number of required drives: 3 drives

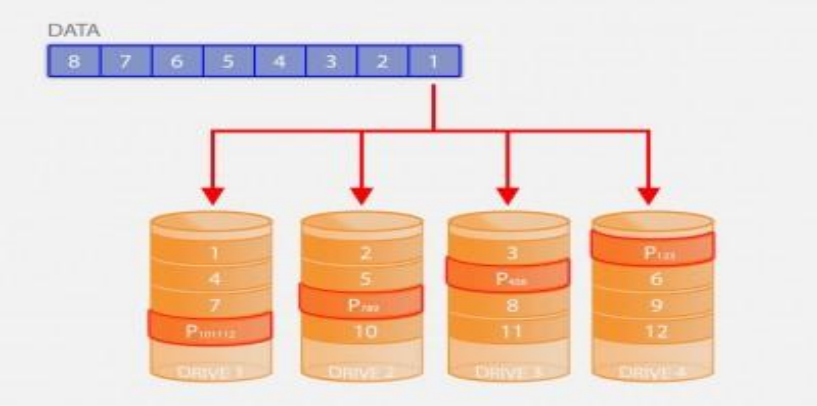
Space efficiency: **Sum of the capacity of all drives – (Sum of the capacity of all drives / number of drives)**

Advantages:

- Good **performance**: Read data transactions are very fast while write data transactions are somewhat slower (due to the parity that has to be calculated).
- Very good data **reliability**: If a drive fails, you still have access to all data, even while the failed drive is being replaced and the storage controller rebuilds the data on the new drive.

Disadvantage:

This is complex technology. If one of the disks in an array using 4TB disks fails and is replaced, restoring the data (the rebuild time) may take a day or longer, depending on the load on the array and the speed of the controller. If another disk goes bad during that time, data are lost forever.



RAID 5 – Striping with distributed Parity

Ideal use

RAID 5 is a good all-round system that combines efficient storage with excellent security and decent performance. It is ideal for file and application servers that have a limited number of data drives.

RAID 6 – Striping and Double rotating Parity

RAID 6 can be seen as an extended RAID 5. It actually maintains 2 parity calculations which are written to two drive, and like RAID 5 these are equally distributed offering the ability to even recover from 2 failing drives.

Minimum number of required drives: 4 drives.

Space efficiency:

Sum of the capacity of all drives – (2 x Sum of the capacity of all drives / number of drives)

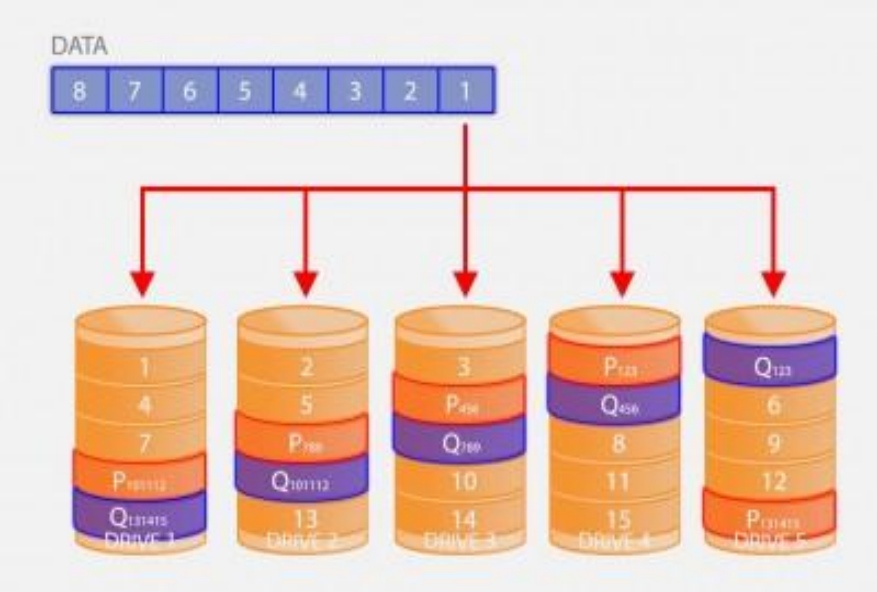
Advantages

- Good **speed** performance: Like with RAID 5, read data transactions are very fast.

- Higher **reliability**: If two drives fail, you still have access to all data, even while the failed drives are being replaced. Therefore, RAID 6 is more secure than RAID5.

Disadvantages

- Write data transactions are slower than RAID 5 due to the additional parity data that have to be calculated.
- The remaining **capacity** will be lower than RAID 5.



RAID 6 – Striping with double distributed parity

Ideal use RAID 6 is a good all-round system that combines efficient storage with excellent security and decent performance. It is preferable over RAID 5 in file and application servers that use many large drives for data storage.

Nested RAID configurations

Certain RAID variations can be combined, depending on your RAID controller of course, and are referred to as "Nested RAID".

Below an example of **RAID 10** or RAID 1 + 0 … Data is written in parallel (RAID 0) and then written mirrored (RAID 1).

It is possible to combine the advantages (and disadvantages) of RAID 0 and RAID 1 in one single system. This is a nested or hybrid RAID configuration. It provides security by mirroring all data on secondary drives while using striping across each set of drives to speed up data transfers.

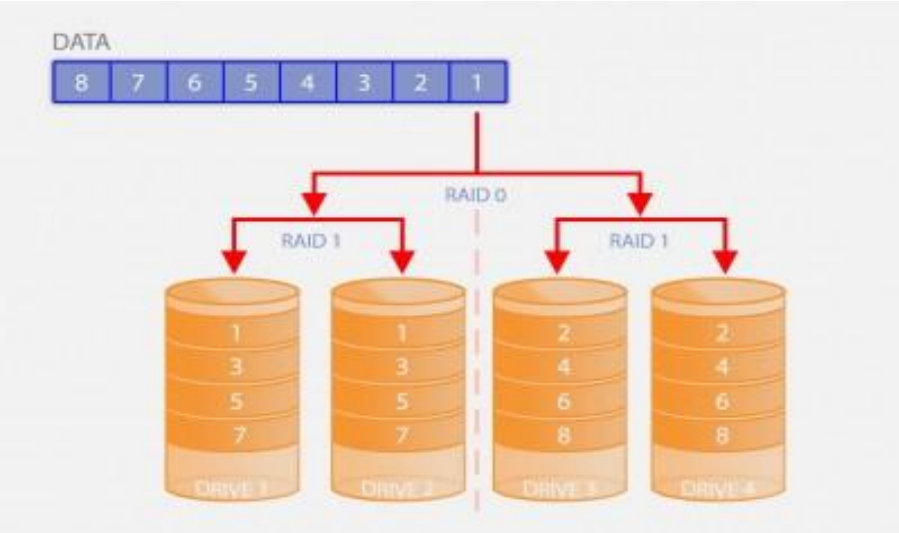
Other variations are **RAID 0+1**, **RAID 50** (RAID 5 + RAID 0), **RAID 100** (RAID 1 + RAID 0 + RAID 0), **RAID 53** (RAID 5 + RAID 3) and **RAID 60** (RAID 6 + RAID 0).

Positives

Custom RAID. You can configure your own fitting your own needs.

Negatives

Not all RAID controllers support this and the resulting **storage capacity** will be much lower than other configurations.



RAID 10 – Nested RAID

JBOD (not RAID)

JBOD, or Just a Bunch Of Disks, is **not a RAID variation**, but most RAID controllers support it.

With JBOD, hard disks are stringed in a sequence, offering the sum of the capacities of all hard disks in the string.

When writing data to the “array”, first disk 1 will be filled and when full, disk 2 will be filled, etc.

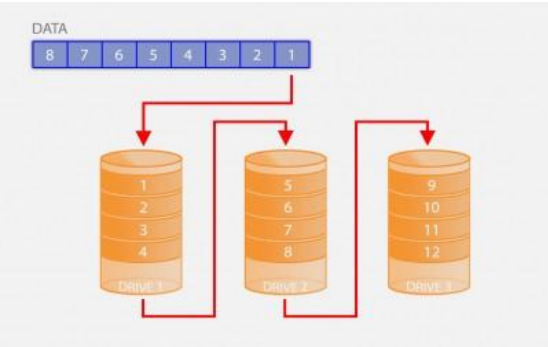
There is no speed gain here, and when a drive fails in a JBOD, not all data is lost, only the data on the defective drive.

For JBOD disks do NOT need to be of the same size.

Positives:

Simple and **Cheap** way of bundling different sized disks.

Negatives: **No increase in speed or reliability.**



JBOD

Distributed Storage and Replication

Introduction

With the popularity of smart phones and sensors, the volume of data being stored and analyzed is growing rapidly, bringing about new business value. Distributed storage is often used to store massive volumes of data, in which multiple hard drives, solid-state drives, and other storage mechanisms are combined to be treated as a single storage. Storage capacity and performance can be improved by adding more servers. In addition, simultaneously storing r**eplicas** of a single data item on multiple servers increases the reliability of data.

Distributed Storage Definition:

Storing data on a multitude of standard servers (using servers local storage rather than using dedicated storage device), which behave as one storage system although data is distributed between these servers.

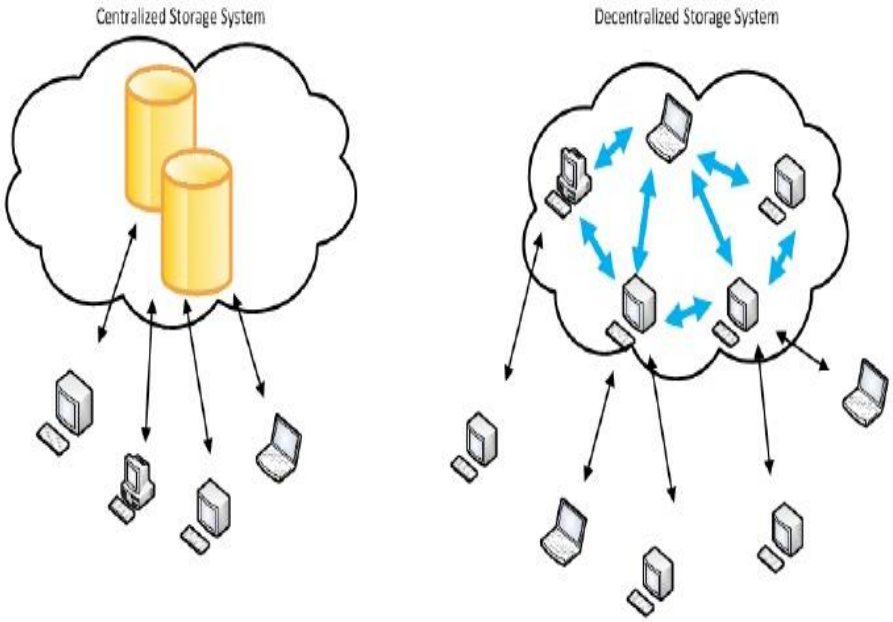
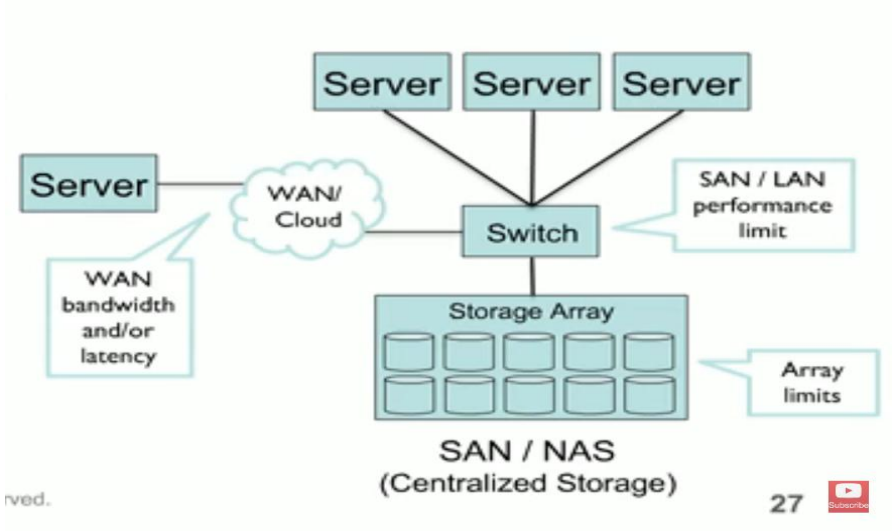
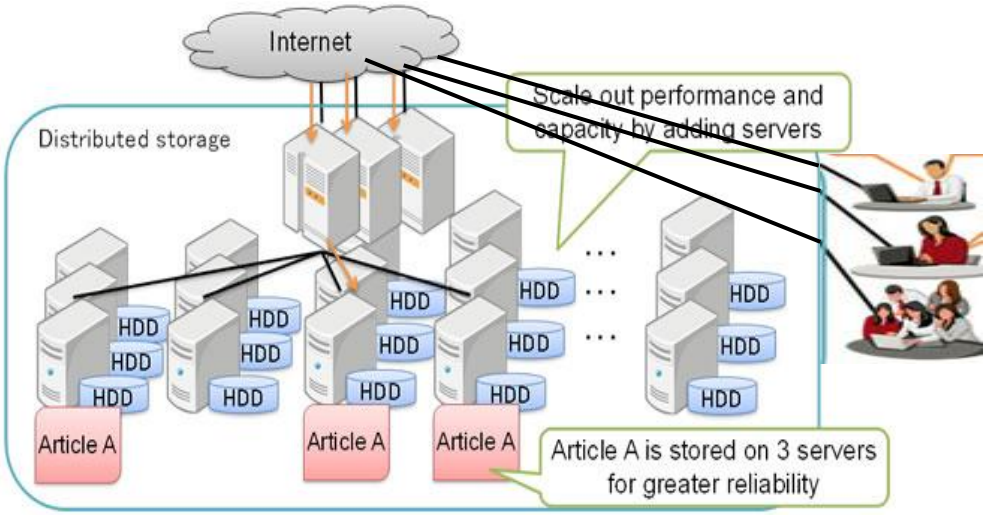


Fig1: Centralized Storage Vs Distributed Storage

Centralized Storage Vs Distributed Storage



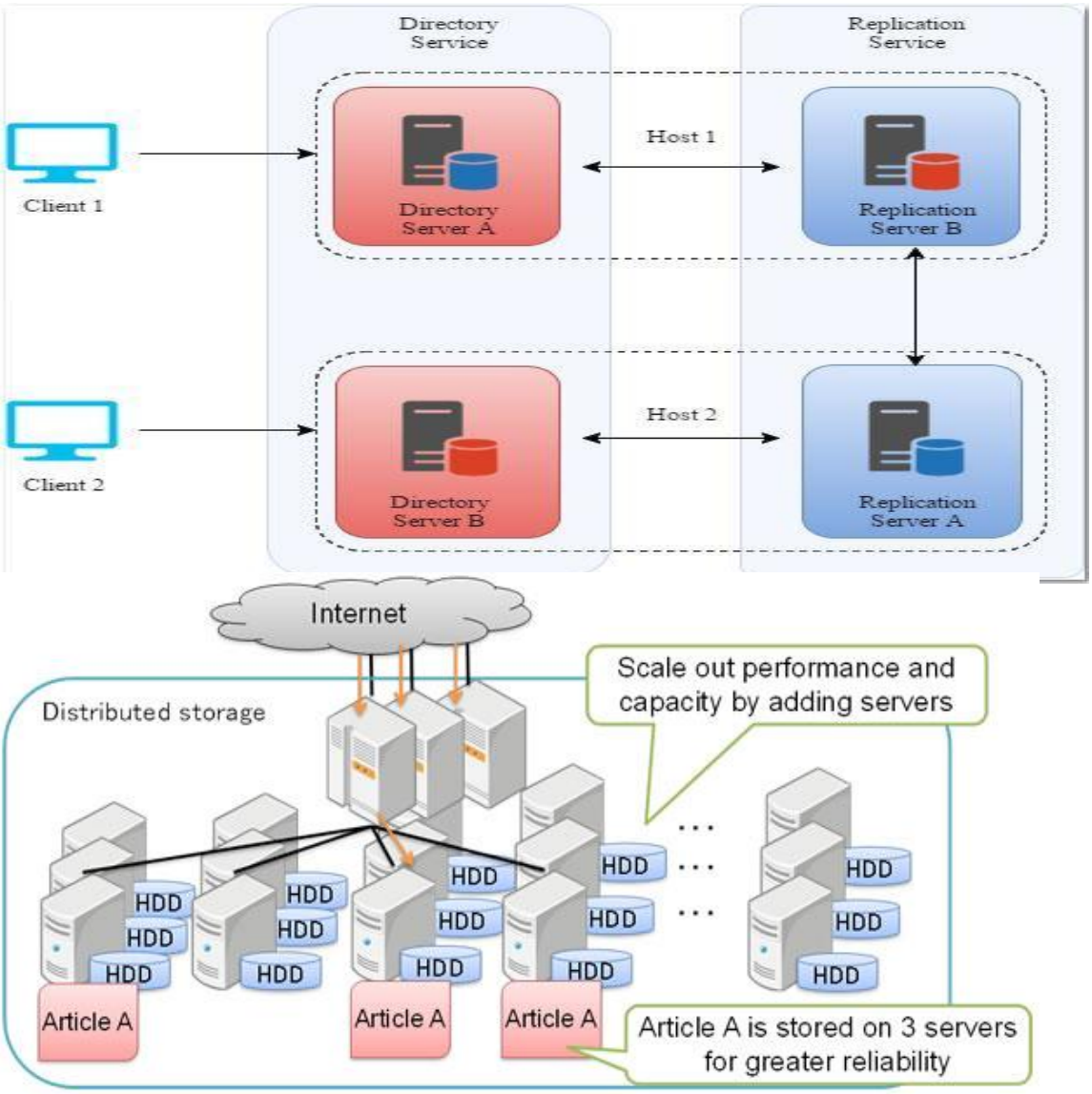
Centralized Storage



Distributed Storage

Replication Mechanism

Replication in Cloud Computing refers to multiple storage of the same data to several different locations by usually synchronization of these data sources. Replication in Cloud Computing is partly done for backup and on the other hand to reduce response times, especially for reading data requests.



Distributed Storage Example

BigTable is a distributed storage system, which is designed by Google, just as its name, it is proposed to deal with data in large scale.

Virtualized Storage

Storage Virtualization

Storage virtualization is **the process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device.**

The process involves **abstracting and covering** the internal functions of a storage device from the host application, host servers or a general network in order to facilitate the application and network-independent management of storage.

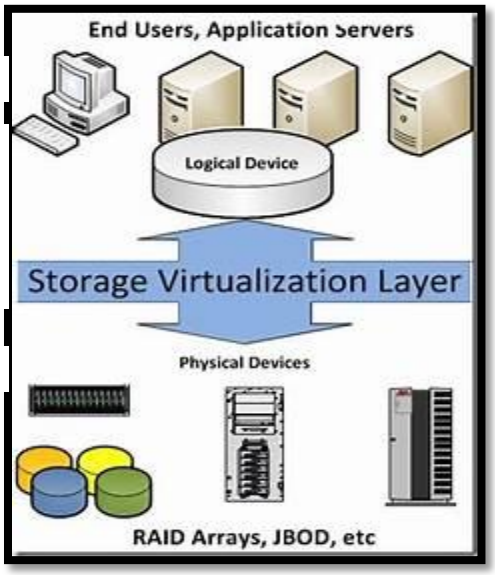
Storage virtualization can be implemented by using software applications or appliances.

The management of storage and data is becoming difficult and time consuming. Therefore, the **storage virtualization** helps to:

1. Improved storage management in a heterogeneous IT environment.
2. Better availability and estimation of down time with automated management
3. Better storage utilization

Storage virtualization can be applied to:

1. Any level of a SAN.
2. A physical storage.
3. RAID groups.
4. Logical unit numbers (LUNs) and logical volumes.



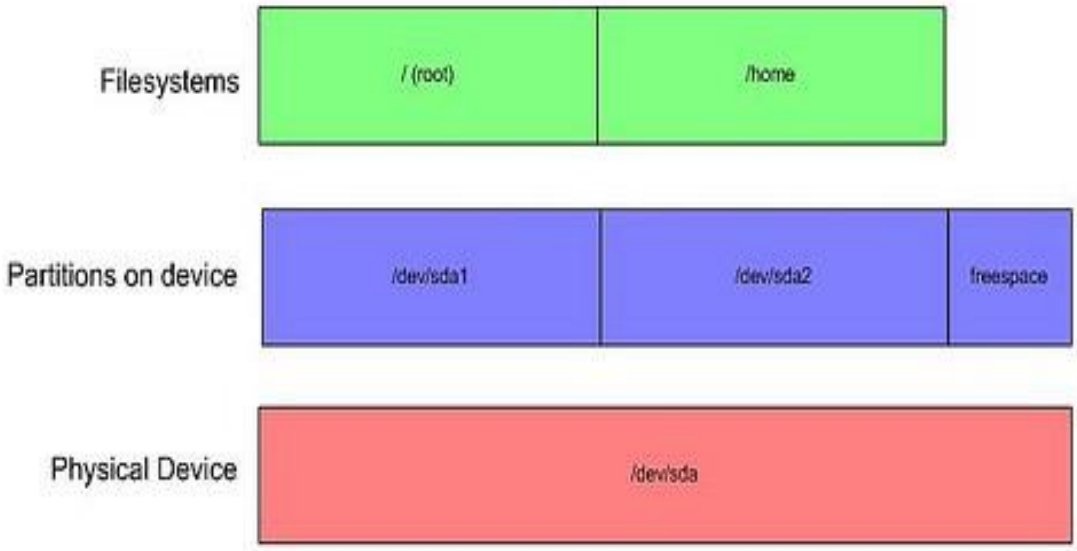
Visualized Storage Creation

We know that **physical disks** reside at the bottom of the storage system, either centralized or distributed. After RAID or replication is implemented, **physical volumes** are created on top of these physical disks.

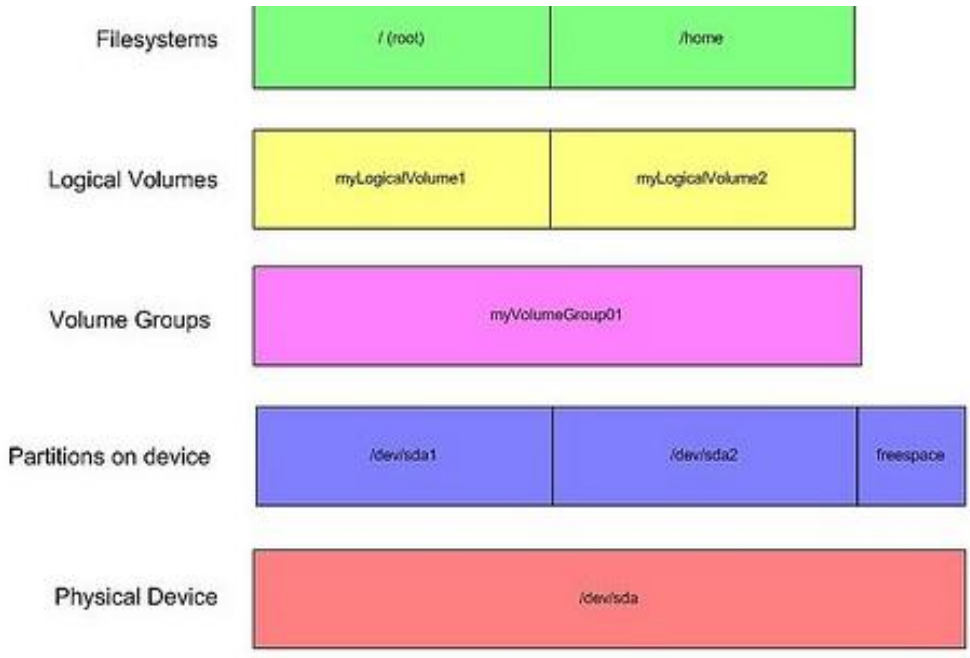
In most cases, the physical volumes are not directly mounted to upper-layer applications, for example, OSs or virtualization systems.

Therefore, typically, multiple physical volumes are combined into a **volume group**, and then the volume group is virtualized into multiple **logical volumes (LVs).** The upper-layer applications use the spaces of the **LVs.**

1. **Traditional Storage System**



2. **Virtualized Storage**



Cloud Storage

Cloud storage Definition

Cloud storage is a cloud computing model in which data is stored on remote servers accessed from the internet, or "cloud." It is maintained, operated and managed by cloud storage service provider on storage servers that are built on virtualization techniques.

Cloud storage works through data center virtualization, providing end users and applications with a virtual storage architecture that is scalable according to application requirements.

Creating Cloud Storage System

The cloud storage system stores multiple copies of data on multiple servers, at multiple locations. If one system fails, then it is required only to change the pointer to the location, where the object is stored.

In cloud computing, the virtualization program formats the LVs.

Vendors use different virtual file systems. For example**, VMware** uses Virtual Machine File System **(VMFS),** and Huawei uses Virtual Image Manage System **(VIMS).** Both of them are high-performance cluster file systems that deliver a capacity exceeding the limit of a single system and allow multiple compute nodes to access an integrated clustered storage pool.

The minimum storage unit used by virtualization programs **is logical unit number (LUN).**

LUNs correspond to volumes. Each LUN represents an individual storage volume (it acts as an identifier for a disk volume in the storage).

Storage Resource Pool: is a pool that pools the physical storage from different network storage devices and makes it appear to be a single storage unit that is handled from a single console.

After virtualization is used, **LUNs can be divided** as:

1. **Thick LUNs**
2. **Thin LUNs.**

	Thick LUNs	**Thin LUNs**
Capacity	- Once being created, get full storage capacity from the storage pool. - Less storage capacity utilization	- Get storage capacity on demand. - Higher storage capacity utilization
Disk space reclamation	**does not apply** to a thick LUN (if data in a thick LUN is deleted, the allocated capacity is occupied by the thick LUN and cannot be used by other LUNs)	When data in a thin LUN is deleted, space in the thin LUN **can be released.**

Performance	A thick LUN delivers **higher performance for sequential** reads/writes as it gets full storage capacity from the beginning, but it has some storage capacity wasted.	A background formatting is required each time the thin LUN expands capacity, In addition, capacity allocations may cause discontinuous disk storage space, so it **takes more time** for sequential reads/writes to find storage locations. Which **cause less performance**.
Application scenarios	− High performance is required. − Storage space utilization is less concerned. − Costs are insensitive.	- Moderate performance is required. − Storage space utilization is more concerned. − Costs are sensitive. − Required storage capacity is hard to predict.

VM Disks

A VM consists of **configuration files and disk files**. Each VM disk corresponds to a disk file where **user data is stored**.

If **virtualized** storage is used, all disk files are stored in the **shared directory** of the file system.

If non-virtualized storage is used, each **disk file corresponds to a LUN**.

From the perspective of users and OSs, either files or LUNs are the same as common hard drives, which are displayed as hard drives among the hardware resources of the system.

When creating a VM, the administrator needs to create disks for the VM to store data. The disk information corresponds to several lines in the configuration file.

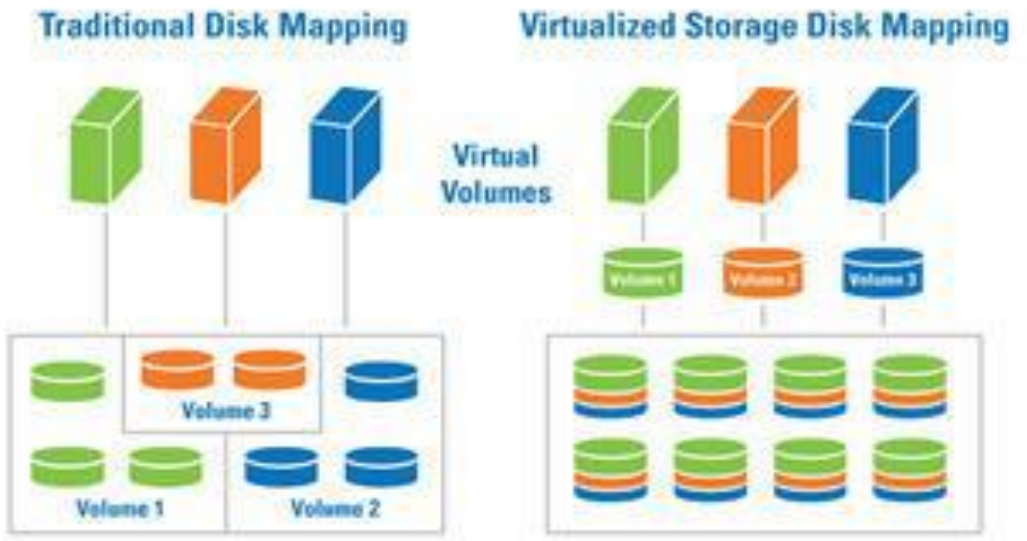
VM disk files have their own fixed formats and there is many common VM disk format as shown in the table:

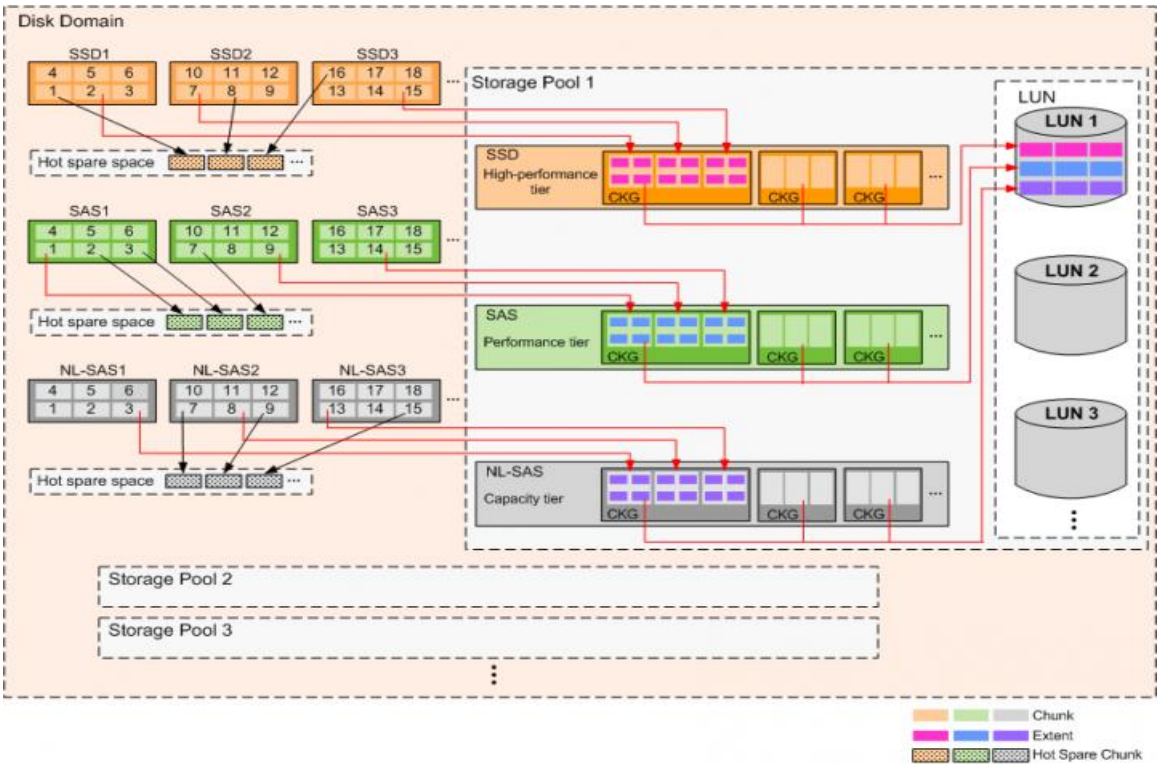
Common VM disk formats

VM Disk File Format	Supported Vendor, Product, or Platform
RAW	All vendors
VMDK	VMware
VHD	Microsoft Hyper-V and Huawei FusionCompute
QCOW QED	QEMU or KVM virtualization platforms
VDI	Oracle

Each vendor can use **its own tool to convert other VM disk formats** to formats that can be used by its own products.

For example, Huawei Rainbow can convert third-party or open-source VM disks to the VHD format.


Traditional Disk Mapping
Virtualized Storage Disk Mapping
Virtual Volumes
Volume 3
Volume 1
Volume 2


Disk Domain
SSD1
SSD2
SSD3
Hot spare space
SAS1
SAS2
SAS3
Hot spare space
NL-SAS1
NL-SAS2
NL-SAS3
Hot spare space
Storage Pool 1
SSD
High-performance tier
SAS
Performance tier
NL-SAS
Capacity tier
CKG
LUN
LUN 1
LUN 2
LUN 3
Storage Pool 2
Storage Pool 3
Chunk
Extent
Hot Spare Chunk

Storage Features of Huawei Virtualization Products

Storage Architecture of Huawei Virtualization Products

1. **Storage Resources:** it is the physical storage media that deliver storage space for virtualization.

   FusionCompute can use the storage resources from dedicated storage devices or the local disks of hosts. Dedicated storage devices are connected to hosts through network cables or optical fibers.

   Storage resources that can be converted to datastores include:

   - LUNs on SAN devices, including iSCSI storage devices and FC SAN storage devices
   - File systems created on network attached storage (NAS) devices
   - Storage pools on FusionStorage Block
   - Local disks on hosts (virtualized)

2. **Storage devices**: In Huawei FusionCompute, these storage units are called storage devices, include: LUNs, File systems, Storage pools, Local disks.

3. **Datastores:** FusionCompute uniformly converts storage resources into **data stores**.

4. **Virtual disks**: After data stores are associated with hosts, **virtual disks** can be created for VMs.



Creating Storage in Huawei

1. Before using datastores, you need to **manually add storage resources**. If the storage resources are IP SAN, FusionStorage, or NAS storage, you need to add storage ports for hosts in the cluster and use the ports to communicate with the service ports of centralized storage controller or the management IP address of FusionStorage Manager. If the storage resources are provided by FC SAN, you do not need to add storage ports.

2. After adding storage resources, you need to scan for these storage devices on the FusionCompute portal to add them as datastores.
3. Datastores can be virtualized or non-virtualized.

4. You can use LUNs as datastores and connect them to VMs from the SAN without creating virtual disks. This process is **called raw device** mapping (RDM). This technology applies to scenarios requiring large disk space, for example, database server construction. RDM can be used only for VMs that run certain OSs.

5. After adding data stores, you can create virtual disks for VMs.

Characteristics of Huawei VM Disks

After adding datastores, you can create virtual disks for VMs. Customers may have various needs for using VMs, for example, they may want to share a VM disk to save more physical space. Therefore, Huawei VM disks are classified into different types based on these requirements.

- **Based on sharing type**, VM disks are classified as non-shared disks and shared disks

  − Non-shared: A non-shared disk can be used only by a single VM.

  − Shared: A shared disk can be used by multiple VMs. If multiple VMs that use a shared disk write data into the disk at the same time, data may be lost. Therefore, you need to use application software to control disk access permissions.

- **Based on the configuration mode,** VM disks can be classified as **common** disks, **thin** provisioning disks, and **thick** provisioning lazy zeroed disks.

  − **Common**: The system allocates disk space based on the disk capacity.
  The performance of the disks in this mode is better than that in the other two modes, but the creation duration may be longer than that required in the other modes.

  − **Thin provisioning:** In this mode, the system allocates part of the configured disk capacity for the first time, and allocates the rest disk capacity based on the storage usage of the disk until the configured disk capacity is allocated. In this mode, datastores can be overcommitted. It is recommended that the datastore overcommit rate not exceed 50%. For example, if the total capacity is 100 GB,

the allocated capacity should be less than or equal to 150 GB. If the allocated capacity is greater than the actual capacity, the disk is in thin provisioning mode.

− **Thick provisioning lazy zeroed**: The system allocates disk space based on the disk capacity. In this mode, the disk creation speed is faster than that in the Common mode, and the I/O performance is between the Common and Thin provisioning modes. This configuration mode supports only virtualized local disks or virtualized SAN storage.

- **Based on the Disk mode,** VM disks can be classified as dependent disks, independent persistent disks, and independent non-persistent disks.

  − **Dependent**: A dependent disk is included in the snapshot. Changes are written to disks immediately and permanently.

  − **Independent persistent**: In this mode, disk changes are immediately and permanently written into the disk, which is not affected by snapshots.

  − **Independent non-persistent**: In this mode, disk changes are discarded after the VM is stopped or restored using a snapshot.

  If you select Independent persistent or Independent non-persistent, the system does not take snapshots of the data on the disk when creating a snapshot for the VM. When the VM snapshot is used to restore a VM, the VM disks are not restored.

  After a snapshot is taken for a VM, if disks on the VM are detached from the VM and not attached to any other VM, the disks will be attached to the VM after the VM is restored using the snapshot.

  However, data on the disks will not be restored. If a disk is deleted after a snapshot is created for the VM, the disk will not be attached to the VM after the VM is restored using the snapshot.

  Some disk types cannot be changed once they are set and some can be changed. For example, **disk modes can be converted.**